

CLINICAL & TECHNICAL SPECIFICATION

Smaran (স্মরণ) — Technical Stack & Architecture Report

Full-Stack Offline-First Cognitive Wellness, Bhashini AI Multilingual Voice & GPS Geofencing System for Dementia & Alzheimer’s Care

PLATFORM	LANGUAGES SUPPORTED	TARGET CONDITION	REPOSITORY / BRANCH
Web, Mobile PWA, Tablet	Assamese, Bodo, English, Hindi	Early to Moderate Dementia	imaritradas/smaran (main)

1. System Architecture Overview

Smaran is engineered as a resilient, dual-portal digital companion comprising an accessible patient interface and a clinical caregiver monitoring console. The system bridges severe infrastructural gaps in elderly dementia management: **offline rural resilience**, **indigenous North-Eastern dialect communication**, and **active wandering prevention**.

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+-----+ | PATIENT
DEVICE (PWA / BROWSER) | | [Daily Activities & Games] [Take Me Home SOS] [Bhashini Multilingual
Voice]| | | | | +-----+ +-----+ +-----+ | | |
IndexedDB Store | | HTML5 Geolocation | | Web Speech Queue | | | (idb v8 - Offline)| | (45s
Throttled) | | (Offline Fallback)| | | +-----+ +-----+ +-----+
-----+ | +-----+ |-----+ |
Background Sync & Rest API v +-----+
-----+ | SMARAN BACKEND ENGINE (Next.js 14) | | - Next.js Route Handlers (/api/bhashini,
/api/location, /api/trends, /api/sync) | | - Mathematical Geofence Engine (Haversine Distance &
Azimuth Bearing) | | - Longitudinal Cognitive Trend Engine (Intake Baseline & 14-Day Rolling
Window) | | - Dual-Mode Storage Layer (Atomic JSON Store & Cloud Firestore) | +-----+
-----+ |-----+ v +-----+
-----+ | CAREGIVER DASHBOARD PORTAL | | [Clinical
Longitudinal Verdict] [Live Geofence Radar] [Prescriptive Plan] | +-----+
-----+
```

2. Core Framework, Language & Runtime

Next.js 14 (App Router & Route Handlers)

FRAMEWORK • V14.2.35

Role in Smaran: Serves as the full-stack backbone. It powers dynamic patient interfaces, static pages, and protected server-side Route Handlers (such as `/api/bhashini/tts` and `/api/location`).

✓ Architectural Rationale & Advantages over Alternatives:

- 1. Absolute Secret Protection:** Pure client-side apps (Vite, Create-React-App) expose API keys in browser network inspection. Next.js Route Handlers guarantee that government Bhashini tokens, Udyat credentials, and Firebase Admin private keys never leave the server.
- 2. Fast Static & Hybrid Optimization:** Delivers pre-compiled lightweight client assets with a shared JS footprint of only 87.3 kB, ensuring zero lag on budget tablets.
- 3. Unified Codebase:** Eliminates the operational complexity, CORS configuration hurdles, and deployment desynchronization of maintaining separate React and Express/Node repositories.

TypeScript 5

LANGUAGE • V5.X

Role: Comprehensive static type safety across data contracts (`types.ts`), GPS coordinates, game metrics, and executive verdicts.

✓ Benefit over JavaScript:

Guarantees compile-time validation. Prevents fatal runtime errors such as undefined coordinates during spatial distance calculations or NaN scores in cognitive evaluations.

React 18

UI RUNTIME • V18.X

Role: Declarative component system powering games, memory recall, and real-time caregiver monitoring.

✓ Benefit over Vue / Angular:

Seamless integration with Next.js Server Components, concurrent rendering hooks, and zero overhead for state updates during SVG radar scanning.

3. Styling, Typography & Dementia Accessibility System

Tailwind CSS + CSS Design Tokens

DESIGN SYSTEM • V3.4.1

Role: Provides utility-first styling with custom CSS variables (`--bg-primary`, `--surface-raised`, `--text-primary`) for instant light and dark theme switching.

✓ Why Used & Advantages over Component Libraries (MUI, AntD, Chakra):

- 1. Zero Runtime JS Overhead:** UI component libraries ship 200KB-400KB of runtime styling logic, creating noticeable input lag for elderly users on mobile browsers. Tailwind compiles down to a compact, static stylesheet.
- 2. Strict Dementia Accessibility (WCAG AAA):** Allows precise configuration of minimum 48x48px touch targets, high-contrast color ratios, large readable typography, and soft glassmorphic visual cues that avoid sensory overload.

Native Vector SVG Visualization (Zero Chart Dependency)

VISUAL ANALYTICS

Role: Handcrafted mathematical SVG graphics powering the dual-ring dials, baseline needles, sparklines, and GPS scanning radar.

✓ Benefit over Chart.js / Recharts / D3.js:

Heavy graphing libraries add 150KB-300KB of bloated JavaScript dependencies and trigger SSR hydration mismatches in Next.js. Custom SVG components render instantaneously, scale to any display resolution, and animate smoothly using hardware-accelerated CSS keyframes.

4. Storage & Offline-First Data Architecture

LAYER	TECHNOLOGY	ARCHITECTURAL RATIONALE & ADVANTAGE
Client Database	IndexedDB via idb (v8.0.3)	Dementia patients frequently travel or reside in remote areas with poor internet. IndexedDB stores sessions, game accuracy, and check-ins asynchronously with structured indexing (by-synced, by-patient). Unlike LocalStorage (5MB cap, synchronous blocking), IndexedDB handles hundreds of megabytes without freezing the UI.
Dual-Mode Server Store	File-backed JSON Store + Cloud Firestore	src/lib/serverStore.ts provides a zero-configuration atomic file database (.data/smaran-db.json) for instant local deployment and hackathons, while seamlessly upgrading to Google Cloud Firestore when cloud credentials are configured.
Cross-Tab Sync	HTML5 BroadcastChannel API	Instantly synchronizes updates (such as caregiver adding a reminder or changing patient language) across all open tabs and windows in real-time without requiring expensive WebSocket servers or battery-draining polling loops.

5. Multilingual AI Voice & Natural Language (Bhashini)

Bhashini ULCA & Dhruva Neural Voice Engine (MeitY, Govt. of India)

INDIGENOUS AI • REST API

Role: Converts text into natural, native speech and recognizes spoken input for North-Eastern languages: Assamese (as), Bodo (brx), English (en), and Hindi (hi).

✓ **Benefit over Google Cloud Speech & AWS Polly:**

1. Authentic North-Eastern Dialects: Western AI platforms lack native conversational models for Bodo and indigenous dialects, producing robotic and misleading pronunciations that disorient dementia patients. Bhashini provides culturally authentic voices.

2. National Sovereign Infrastructure: Built on Digital India public digital infrastructure, respecting privacy and sovereign data protocols.

Web Speech API Fallback & Speech Queue Engine

RESILIENCE LAYER

Role: Located in src/lib/i18n.ts. If Bhashini network latency exceeds 3 seconds or the device is offline, it instantly routes speech through browser speech synthesis with zero UI hang.

✓ **Solved Engineering Challenges:**

Implements an in-memory speech FIFO queue to prevent overlapping voices, handles Chrome's known 15-second speech synthesis pause bug, and caches base64 audio to minimize network data consumption.

6. Geolocation, Safe-Zone Geofencing & 'Take Me Home' SOS

HTML5 Geolocation API

SPATIAL API

Implementation: PatientGPSBeacon.tsx

uses navigator.geolocation.watchPosition with high accuracy and a 45-second throttling window.

✓ Advantage:

Eliminates the requirement for families to purchase expensive proprietary GPS hardware trackers; functions on any smartphone the patient carries.

Haversine & Azimuth Engine

SPHERICAL MATH

Implementation: Pure mathematical computation in serverStore.ts for spherical distance and forward azimuth degrees (0°-360°).

✓ Advantage:

Instant zero-latency boundary breach detection and real-time rotating compass navigation with zero Google Maps API costs.

Clinical Wandering Prevention & Emergency Assistance Flow

- Automatic Geofence Breach:** If distance to home exceeds the configured safe radius (100m - 2000m), the server automatically dispatches an immediate high-priority wandering advisory alert.
- Compassionate "Take Me Home" Modal:** When a confused patient presses the emergency button, Bhashini speaks reassuring words in their native tongue: *"Don't worry, we are helping you get home. Your family has been notified with your exact location."*
- Visual Orientation:** Displays a high-contrast compass pointer oriented towards home, distance in meters, and a 1-tap phone dialer to call the caregiver.

7. Clinical Cognitive Domain Modeling & Verdict

COGNITIVE DOMAIN	SMARAN ACTIVITY	CLINICAL DIAGNOSTIC PURPOSE
Short-Term Memory	Memory Match (Visual card pairing)	Assesses hippocampal recall retention and visual memory deterioration.
Visuomotor Attention	Spot and Tap (Target detection)	Monitors reaction latency (ms), visual discrimination, and motor speed.
Pattern Recognition	Sequence Recall (Forward color span)	Evaluates prefrontal working memory capacity and structured logic.
Procedural Memory	Routine Recall (Daily task ordering)	Measures ability to order activities of daily living (brushing, breakfast, medicine).

Longitudinal Assessment & 14-Day Rolling Baseline

Unlike single-instance tests which are skewed by acute fatigue or dehydration, `src/lib/trends.ts` computes an **Intake Baseline** and evaluates a **14-day rolling window**. It automatically outputs a unified **Executive Cognitive Verdict** with risk categorization (*Optimal, Stable, Moderate Risk, Elevated Risk*) and prescriptive care plan recommendations.

8. Complete Technology Stack Matrix

COMPONENT	TECHNOLOGY	VERSION	KEY DIFFERENTIATING BENEFIT
Framework	Next.js (App Router)	14.2.35	Server-side secret isolation, unified fullstack repo, fast PWA delivery
Language	TypeScript	5.x	Guaranteed type safety across medical metrics & spatial coordinates
Styling	Tailwind CSS	3.4.1	Near-zero runtime styling overhead; WCAG AAA dementia accessibility
Data Visuals	Custom Vector SVGs	Native	Zero chart library bundle bloat; hardware-accelerated fluid rendering
Offline Store	IndexedDB (via idb)	8.0.3	Unlimited asynchronous storage; full gameplay without network access
Server Store	Dual-Store (JSON + Firestore)	v12 / v14	Zero-config file persistence with seamless cloud enterprise upgrade
Cross-Tab Sync	BroadcastChannel API	Native	Instant zero-battery cross-window synchronization without WebSockets
Regional AI Voice	Bhashini Dhruva / ULCA	v0 Pipeline	Indigenous North-Eastern language voices (Assamese, Bodo, Hindi)
Speech Fallback	Web Speech API + FIFO Queue	Native	Zero voice overlap; handles Chrome 15s utterance pause bug smoothly
Location Sensor	HTML5 Geolocation	Native	Battery-friendly 45s throttled position tracking on any smartphone
Geofence Math	Haversine & Azimuth Engine	Native	Millisecond spatial boundary checking with zero Google Maps API costs